Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Following the Order of Operations in Algebra

When you simplify an algebraic expression, you must deal with its parts in a fixed order. Just as with numbers, the rule is: work out anything inside brackets first, then deal with powers, then do multiplication and division from left to right, and finally do addition and subtraction from left to right. A handy way to remember the order is BODMAS: Brackets, Of (or powers), Division, Multiplication, Addition, Subtraction.

The letters stand for:

Brackets – clear any brackets first.
Order / Of – handle powers like squares and cubes.
Division and Multiplication – do these from left to right as they appear.
Addition and Subtraction – do these from left to right last.

The same order works whether the letters are numbers or symbols, so an expression like 2x + 3 × x is not simplified from left to right blindly – the multiplication 3 × x = 3x comes before the addition, giving 2x + 3x = 5x. Keep this order in mind and every expression will come out the same way every time.

Worked Example 1 – no brackets
Simplify 4x + 2 × 3x.
Step 1: Do the multiplication first. 2 × 3x = 6x.
Step 2: Now add. 4x + 6x = 10x.
Answer: 10x

Worked Example 2 – with brackets
Simplify 3(2x + 4).
Step 1: Deal with the bracket first by multiplying each term inside by the number outside. 3 × 2x = 6x and 3 × 4 = 12.
Step 2: Write the result: 6x + 12.
Answer: 6x + 12

Worked Example 3 – brackets inside a longer expression
Simplify 2x + 4(3x − 5).
Step 1: Multiply out the bracket first. 4 × 3x = 12x and 4 × (−5) = −20, so the bracket becomes 12x − 20.
Step 2: Bring the rest down: 2x + 12x − 20.
Step 3: Collect like terms. 2x + 12x = 14x, so we get 14x − 20.
Answer: 14x − 20

Let's see how the order of operations applies when you have several different steps in one expression.

Worked Example 4 – multiplication and addition together
Simplify 5x − 2 × 3x + 4.
Step 1: Multiplication first. 2 × 3x = 6x.
Step 2: Replace and simplify from left to right: 5x − 6x + 4.
Step 3: 5x − 6x = −1x, or −x, so we have −x + 4.
Answer: −x + 4

Worked Example 5 – a bracket containing two terms
Simplify 2(4x + 3) − 5x.
Step 1: Multiply out the bracket: 2 × 4x = 8x and 2 × 3 = 6, giving 8x + 6.
Step 2: Bring down the rest: 8x + 6 − 5x.
Step 3: Collect like terms: 8x − 5x = 3x, so we have 3x + 6.
Answer: 3x + 6

When a negative sign sits in front of a bracket, it multiplies every term inside by −1. For example, 3x − (2x + 1) becomes 3x − 2x − 1 = x − 1. Take care to change the sign of each term inside.

Worked Example 6 – a bracket with a minus in front
Simplify 4x − (2x + 3).
Step 1: The minus in front means −1 × (2x + 3) = −2x − 3.
Step 2: Bring down the front term: 4x − 2x − 3.
Step 3: 4x − 2x = 2x, so we get 2x − 3.
Answer: 2x − 3

Practice

1. Simplify 3x + 2 × 4x.
2. Simplify 5(2x + 3).
3. Simplify 4x + 3(2x − 1).
4. Simplify 6x − (4x + 2).

Section 2: Working with Brackets in Algebra

In algebra, a bracket groups terms together and tells you to treat everything inside it as a single unit. Two skills matter here: taking a bracket away (removing or expanding it) and putting one in (factorising or inserting it). Both rely on one idea — the sign that sits in front of the bracket decides what happens to every sign inside.

Removing a bracket (expanding). The term just in front of the bracket multiplies every term inside, one by one. Watch the sign in front:

• If the sign in front is +, the signs inside stay exactly as they are.
• If the sign in front is −, every sign inside flips: + becomes − and − becomes +.
• If a number or term stands in front (like 3 or 2x), multiply that with each term inside as well as applying the sign rule.

For example, to remove the bracket in 3(a + b): multiply 3 by a and by b, keeping the +, to get 3a + 3b. To remove the bracket in −(c − d): the minus flips the signs, so −c + d.

Putting a bracket in (inserting). You can gather terms that share a common factor and write the common factor in front of a bracket. Find the biggest factor common to every term, pull it out front, and divide each term by it to see what goes inside.

For example, 4x + 4y shares a factor of 4, so it becomes 4(x + y). And −6a + 6b shares a factor of 6 (the signs are different, so the bracket carries the −): −6a + 6b = −6(a − b).

Always simplify after removing or inserting a bracket. Simplification means combining like terms — terms with exactly the same letters and powers — by adding or subtracting their number parts (their coefficients). Terms that do not match cannot be combined.

Here is the full sequence to follow for any expression with brackets:
1. Remove any brackets, applying the sign and multiplication rules.
2. Collect the like terms together.
3. Add or subtract the coefficients of like terms.
4. Write the final answer neatly, with the terms in a sensible order.

When you insert a bracket, the same idea runs backwards: pick out a common factor and group what is left inside. Your answer should always be checked — expand your bracket back out to confirm you get the original expression. If inserting and removing give you back what you started with, your algebra is correct.

Worked example 1 — removing a bracket with a plus sign in front

Simplify: 5 + (2x + 3)

Step 1: The bracket has + in front, so the signs inside stay. Remove it: 5 + 2x + 3.
Step 2: Collect the like terms. The numbers 5 and 3 are like terms: 5 + 3 = 8.
Step 3: Bring the 2x down: 2x + 8.

Answer: 2x + 8.

Worked example 2 — removing a bracket with a minus sign in front

Simplify: 4y − (2y + 1)

Step 1: The bracket has − in front, so every sign inside flips. Remove it: 4y − 2y − 1.
Step 2: Collect like terms. The y-terms match: 4y − 2y = 2y.
Step 3: Bring the −1 down: 2y − 1.

Answer: 2y − 1.

Worked example 3 — multiplying a bracket by a number and a letter

Simplify: 3(2a + 1) + 2(a − 4)

Step 1: Expand each bracket separately.
3(2a + 1) = 3 × 2a + 3 × 1 = 6a + 3
2(a − 4) = 2 × a − 2 × 4 = 2a − 8
Step 2: Combine the pieces: 6a + 3 + 2a − 8.
Step 3: Collect like terms. The a-terms: 6a + 2a = 8a. The numbers: 3 − 8 = −5.
Step 4: Write the answer: 8a − 5.

Answer: 8a − 5.

Worked example 4 — inserting a bracket (factorising)

Simplify by inserting a bracket: 6m + 9

Step 1: Find the biggest factor common to 6m and 9. The factors of 6m are 1, 2, 3, 6 and m; the factors of 9 are 1, 3, 9. The common factor is 3.
Step 2: Divide each term by 3: 6m ÷ 3 = 2m, and 9 ÷ 3 = 3.
Step 3: Write the common factor in front of the bracket with what is left inside: 3(2m + 3).

Check: expand 3(2m + 3) = 6m + 9. ✓

Answer: 3(2m + 3).

Worked example 5 — inserting a bracket when the signs differ

Simplify by inserting a bracket: 8p − 12

Step 1: The common factor of 8p and 12 is 4.
Step 2: Divide each term by 4: 8p ÷ 4 = 2p, and 12 ÷ 4 = 3. Because the signs differ, put the minus inside the bracket.
Step 3: Write: 4(2p − 3).

Check: expand 4(2p − 3) = 8p − 12. ✓

Answer: 4(2p − 3).

Practice

1. Simplify: 7 + (3x + 2)
2. Simplify: 5a − (2a + 4)
3. Simplify: 2(3b + 1) + 4(b − 2)
4. Insert a bracket to simplify: 12m + 8
5. Insert a bracket to simplify: 10n − 15

Section 3: From Words to Algebra: Solving Word Problems

A word problem is a story that hides a number question inside it. To solve it, you read the story, choose a letter for the unknown amount, and then use the algebra rules you already know to turn the sentences into one clean expression or equation.

Work through these three steps every time:
1. Choose a letter for the unknown. Write down what it stands for, for example "Let x be the number of mangoes."
2. Turn each sentence into a piece of algebra. Watch for the key words: "more than" means add, "less than" means subtract, "twice" and "double" mean multiply by 2, and "three times as many" means multiply by 3.
3. Join the pieces into one expression, simplify it (collect like terms), and if you can, solve it to find the number.

The important skill here is translating words into algebraic terms, then simplifying what you get, exactly like you have been doing with expressions.

Example 1: Tunde has t books. His sister has 5 more books than Tunde, and his friend has twice as many books as Tunde. Write a simplified expression for the total number of books.

Step 1: Let t be the number of books Tunde has.
Step 2: Turn each sentence into algebra.
- Sister has 5 more than Tunde: $t + 5$.
- Friend has twice as many as Tunde: $2t$.
Step 3: Add the pieces for the total: $t + (t + 5) + 2t$.
Simplify by collecting like terms: $t + t + 2t = 4t$, and keep the $+ 5$.
Total $= 4t + 5$ books.

Example 2: A number is multiplied by 3 and then 4 is added, and the result is 22. Find the number.

Let the number be n.
"Multiplied by 3": $3n$. "Then 4 is added": $3n + 4$. "The result is 22": $3n + 4 = 22$.
Subtract 4 from both sides: $3n = 18$.
Divide both sides by 3: $n = 6$.
The number is 6. Check: $3 \times 6 + 4 = 18 + 4 = 22$. Correct.

Practice

1. Adaeze has a oranges. Her friend has 3 fewer oranges than Adaeze, and her brother has 4 times as many oranges as Adaeze. Write a simplified expression for the total number of oranges they all have together.
2. A number is doubled and then 7 is subtracted from the result, giving 15. Find the number.
3. A rectangle has a length that is 6 cm more than its width, w. Write and simplify an expression for the perimeter of the rectangle.
4. Chinedu is m years old. His father is 25 years older than Chinedu, and his younger sister is 10 years younger than Chinedu. Write a simplified expression for the sum of their three ages.

Check your work

Attempt every question before you read this page.

Following the Order of Operations in Algebra

1. $11x$

   Do the multiplication first: $2 \times 4x = 8x$. Then add: $3x + 8x = 11x$.
2. $10x + 15$

   Multiply each term inside the bracket by 5: $5 \times 2x = 10x$ and $5 \times 3 = 15$, giving $10x + 15$.
3. $10x - 3$

   Multiply out the bracket first: $3 \times 2x = 6x$ and $3 \times (-1) = -3$, so the bracket becomes $6x - 3$. Bring down the front: $4x + 6x - 3 = 10x - 3$.
4. $2x - 2$

The minus in front means $-1 \times (4x + 2) = -4x - 2$. Bring down the front: $6x - 4x - 2 = 2x - 2$.

Working with Brackets in Algebra

1. $3x + 9$

   The bracket has + in front, so signs stay: $7 + 3x + 2$. Collect the numbers $7 + 2 = 9$, keep $3x$. Answer: $3x + 9$.
2. $3a - 4$

   The bracket has − in front, so signs flip: $5a - 2a - 4$. The a-terms: $5a - 2a = 3a$. Bring -4 down. Answer: $3a - 4$.
3. $10b - 6$

   Expand: $2(3b + 1) = 6b + 2$, and $4(b - 2) = 4b - 8$. Combine: $6b + 2 + 4b - 8$. b-terms: $6b + 4b = 10b$. Numbers: $2 - 8 = -6$. Answer: $10b - 6$.
4. $4(3m + 2)$

   Common factor of $12m$ and 8 is 4. Divide: $12m \div 4 = 3m$, $8 \div 4 = 2$. Write: $4(3m + 2)$. Check: $4 \times 3m + 4 \times 2 = 12m + 8$. ✓
5. $5(2n - 3)$

   Common factor of $10n$ and 15 is 5. Divide: $10n \div 5 = 2n$, $15 \div 5 = 3$. Signs differ, so minus inside: $5(2n - 3)$. Check: $5 \times 2n - 5 \times 3 = 10n - 15$. ✓

From Words to Algebra: Solving Word Problems

1. Let a be the number of oranges Adaeze has. Friend has $a - 3$, brother has $4a$. Total $= a + (a - 3) + 4a = 6a - 3$ oranges.

   Let a be Adaeze's oranges. Friend: $a - 3$. Brother: $4a$. Total $= a + (a - 3) + 4a$. Collect like terms: $a + a + 4a = 6a$, keep -3. So the total is $6a - 3$ oranges.
2. $n = 11$. Check: $2 \times 11 - 7 = 22 - 7 = 15$.

   Let the number be n. Doubled: $2n$. Subtract 7: $2n - 7$. This equals 15, so $2n - 7 = 15$. Add 7 to both sides: $2n = 22$. Divide by 2: $n = 11$. Check: $2(11) - 7 = 22 - 7 = 15$. Correct.
3. Length $= w + 6$. Perimeter $= 2(w + 6) + 2w = 4w + 12$ cm.

   Width $= w$. Length $= w + 6$. Perimeter is the sum of all four sides: $w + (w + 6) + w + (w + 6) = 4w + 12$ cm.
4. $m + (m + 25) + (m - 10) = 3m + 15$ years.

   Chinedu: m. Father: $m + 25$. Sister: $m - 10$. Sum $= m + (m + 25) + (m - 10)$. Collect like terms: $m + m + m = 3m$; and $25 - 10 = 15$. So the sum is $3m + 15$ years.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-ba272936	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____